

FROM DATA TO DIAGNOSIS: SYSTEMATIC AND BIBLIOMETRIC ANALYSIS OF ML-BASED WEARABLE CARDIAC MONITORING

Summer Internship Project report submitted in partial fulfilment of the requirement for MBA program of IFHE

by

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DECLARATION

I hereby declare that this SIP report submitted to IBS Bangalore, the Off-campus centre of IFHE, in partial fulfilment of the requirements of MBA program, is original and is developed by me. I hereby declare that the project report titled, “FROM DATA TO DIAGNOSIS: SYSTEMATIC AND BIBLIOMETRIC ANALYSIS OF ML-BASED WEARABLE CARDIAC MONITORING” is written and submitted by me under the guidance of Dr. Roshny Unnikrishnan and Mrs. Sadhvi Gurudatha. I am very thankful to both for helping me complete project successfully.

I hereby confirm the originality of the work and that there is no plagiarism in any part of the report.

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CERTIFICATE

This is to certify that the thesis submitted by GUNJAN SAHOO (24BSOCBL0572), titled “FROM DATA TO DIAGNOSIS: SYSTEMATIC AND BIBLIOMETRIC ANALYSIS OF ML-BASED WEARABLE CARDIAC MONITORING” is a record of the summer internship project carried out during the February to May 2025 under our supervision.

This SIP report has not formed the basis for the award of any degree, and we hereby confirm the originality of the work and that there is no plagiarism in any part of the report.

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Mrs. Sadhvi Gurudatha

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to all those who have contributed to successfully completing this Final Report.

First and foremost, I would like to express my gratitude to Technomers for giving me the chance to work as an intern at their prestigious company. My awareness of the corporate world and my exposure to the analytic field has greatly helped me and my abilities have been sharpened and experience I have gained while working at Proficient Infosoft will benefit me a lot.

I am also grateful to Mrs Sadhvi Gurudatha , who oversaw my internship and provided invaluable mentorship, support, and advice. I also would like to thank my Guide Dr. Roshny Unnikrishnan who guided me at every step in my internship and my institution ICFAI Business School, Bangalore.

Finally, I would like to sincerely thank everyone who has helped with this internship report, whether directly or indirectly.

GUNJAN SAHOO

EXECUTIVE SUMMARY

I Gunjan Sahoo (Enrolment No.: 24BSOCBL0572) undertook a summer internship at Technomers, the project was titled “**From Data to Diagnosis: Systematic and Bibliometric Analysis of ML-Based Wearable Cardiac Monitoring**”. The firm operated in the IT and software consulting industry. Technomers is known for delivering cutting-edge technological solutions, with expertise in software development, .NET, data science, and AI/machine learning. The organization serves a wide range of sectors, including cloud services, BFSI, robotics, healthcare, and retail, with a strong focus on technical excellence and innovation.

The primary objective of the internship was evaluating the performance of machine learning techniques—specifically Random Forest and XGBoost—for personalized activity type detection using the WESAD dataset. Which was supported by a systematic literature review (SLR), conducted to understand current research trends in machine learning applications for cardiac disease detection and activity classification. The literature review identified Random Forest as one of the most accurate and commonly used model. Building upon these insights, the project involved hands-on application of the identified techniques to validate their effectiveness.

The methodology followed in the project included comprehensive data cleaning and preprocessing of both Scopus and WESAD datasets. Trends from the SLR were analyzed and visualized through bibliometric graphs to identify key focus areas in existing research. These findings informed the development of machine learning models tailored for activity detection. Finally, the performance of the models was evaluated to assess their accuracy and relevance to real-world health monitoring use cases.

XGBoost demonstrated superior performance in terms of accuracy and consistency compared to other models such as SVM and Random Forest. Additionally, exploratory analysis—including respiration signal processing and recurrence plot analysis—offered meaningful insights into physiological patterns associated with different activity types.

The overall conclusion of the project emphasizes the importance of delivering **data-driven, client-centric healthcare solutions**. By carefully analyzing existing literature and identifying dominant machine learning trends, the project aimed to provide Technomers and its clients with a strong, evidence-based recommendation to improve digital health interventions.

To further enhance the effectiveness of the proposed models, it is recommended to perform **hyperparameter tuning** and **cross-validation** to improve performance metrics. Future work should also explore more advanced techniques such as **LSTM or CNN**, which are better suited for handling time-series and biosignal data. Moreover, involving **healthcare professionals** in the model evaluation process can help ensure medical relevance and real-world applicability.

One of the key achievements was the identification and recommendation of optimal machine learning models, specifically Random Forest and XGBoost, for use in wearable health monitoring solutions. Gunjan also developed a structured methodology that integrated bibliometric analysis, data modeling, and visualization—offering a reusable framework for future projects in the health-tech domain. Additionally, the creation of Python scripts and analytical workflows for signal processing and feature extraction from physiological datasets added significant technical value to the company’s toolkit. These contributions also played a role in supporting data-driven decision-making for Technomers’ client, Movano, by assessing the feasibility and impact of machine learning-based cardiac monitoring systems.

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LIST OF SYMBOLS AND ABBREVIATIONS

Empatica E4 and Respiban data

Abbreviation	Full Form
--------------	-----------

ECG	Electrocardiogram
EDA	Electrodermal Activity
TEMP	Temperature Data
BVP	Photoplethysmograph Data
ACC	3-Axis Accelerometer Data
IBI	Inter-Beat Interval Data
HR	Heart Rate Data

RESPIRATORY (Respiration signal, no abbreviation)

Other Abbreviations

Abbreviation	Full Form
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CVD	Cardiovascular Disease
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
ML	Machine Learning
SVM	Support Vector Machine

CHAPTER 1

INTRODUCTION

Technomers, headquartered in Pakshikere, Mangaluru, Karnataka, India, is a dynamic product engineering and consulting company that delivers comprehensive technology solutions to a global clientele. With a strategic presence extending to the USA and Singapore, they specialize in cutting-edge services encompassing software development, .NET expertise, data science, and AI/machine learning, catering to diverse sectors like cloud services, BFSI, robotics, healthcare, and retail. Emphasizing technical excellence and innovative solutions, Technomers is committed to providing best-practice-driven results, enabling businesses to thrive in the rapidly evolving digital landscape.

They provide cutting-edge product engineering and consulting from Mangaluru to the world, specializing in advanced technologies across multiple domains. They work with clients across various industries and verticals. They enable clients across India, the USA, and Singapore to leverage cutting-edge technologies. To mention a few of the verticals of their clients work in are Cloud Services, BFSI, Robotics, Clean Energy, Healthcare, VoIP, WebRTC, and Retail. Client success stories, delivering tangible value across industries and technologies to mention a few are HDFC BANK - fintech, HDFC ERGO-digital transformation, ICICI PRUDENTIAL-insurance tech, NTTDATA-consulting, MAHINDRA-.NET digital transformation, MOVANO- smart devices data science, BREVERA- embedded, DRONYAERIALS- IOT, TRASHTRAP- robotics, KHASSO- retail, TETHERFI- ax cx, NOVIGO- consulting.

As a Business Analyst Intern at Technomers, I'm currently involved in a strategic initiative focused on identifying optimal technological solutions for health data analytics through wearable technologies using machine learning to detect heart disease. This article reviewed multiple research papers to determine the most optimal machine learning technique that can be used to detect heart disease. The intent is to synthesize complex data into clear, actionable insights.

Machine learning techniques such as Random Forest and XGBoost, along with data visualization and a systematic literature review, were employed to demonstrate the success rate of the recommended machine learning methods, complemented by exploratory analysis. The exploratory

analysis utilized data obtained from the UCI Machine Learning Repository. Using this dataset, respiration analysis and recurrence analysis were performed. The data had been collected using the RespiBan device. To gain practical experience in applying machine learning for personalized activity type detection, the Random Forest method—identified as effective through the literature review—was used to train and test the model.

1.1 EIC ANALYSIS

The E-I-C (Economy-Industry-Company) framework provides a structured approach to analyze investment opportunities by examining economic trends, industry dynamics, and company-specific factors. This comprehensive analysis helps investors to make informed decisions by understanding the broader economic environment, the specific industry's characteristics, and the company's potential within that context.

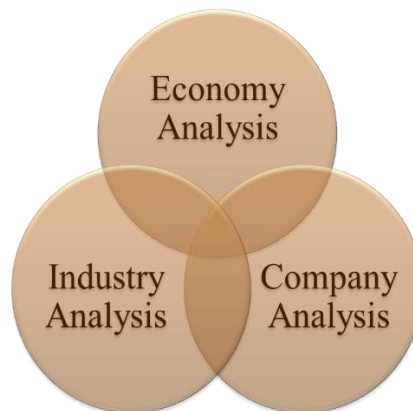


Fig 1.1 EIC ANALYSIS

1.1.1 Economy Analysis

Economic analysis involves understanding the macroeconomic environment in which industries and companies operate. Key aspects of this analysis include:

- **Growth Rates of National Income:** Monitoring Gross National Product (GNP), Net National Product (NNP), and Gross Domestic Product (GDP) to gauge the economy's overall health.
- **Inflation:** Understanding how the Consumer Price Index (CPI) and Wholesale Price Index (WPI) impact consumer purchasing power and business costs.

- **Interest Rates:** Analyzing how interest rates affect the cost and availability of credit, influencing investment and profitability.

Technomers, located in the burgeoning IT hub of Mangalore, likely experienced a period of growth aligned with the national and regional IT sector. India's IT industry saw a robust expansion, contributing significantly to the GDP, with revenue reaching approximately \$227 billion in FY22 and an estimated \$245 billion in FY23. While specific financial figures for Technomers are unavailable, Mangalore itself is becoming an increasingly important IT destination, attracting investments and talent due to its strategic location, cost efficiency compared to larger cities, and a growing skilled workforce. This favorable ecosystem, supported by government initiatives to promote IT in tier-2 cities, would have provided a positive backdrop for Technomers' operations in product engineering, software development, and emerging technologies like AI and data science. However, as an unfunded company competing with larger players, their growth would have been organic and potentially more constrained compared to their funded counterparts in this rapidly expanding market.

1.1.2 Industry Analysis

Industry Background

The IT industry accounted for 7.5% of India's GDP, as of FY23 and is expected to rise to 10% by FY25. According to the National Association of Software and Service Companies (NASSCOM), the Indian IT industry's revenue touched US\$ 227 billion in FY22, a 15.5% YoY growth. IT spending in India is estimated to record double-digit growth of 11.1 % in 2024, totaling US\$ 138.6 billion up from US\$ 124.7 billion last year. Artificial Intelligence (AI) is seen as the economic multiplier of the century, projected to inject US\$ 500 billion into India's GDP by 2025. According to a report published by Microsoft, consultancy firm Bain & Co, and industry body Internet and Mobile Association of India (IAMAI), India produces 16% of the entire world's AI talent pool – the third highest in the world. PwC India also announced that it was planning to hire 10,000 employees in the cloud and digital technologies space over the next five years.

Technomers, established in 2010 and based in Mangaluru, operates within the city's rapidly expanding IT services sector, which has seen a 20% increase in job creation in the last two years

and is home to over 55 companies. While Technomers is currently unfunded and competes with larger entities, it strategically focuses on product engineering, software development, and emerging technologies like AI and data science, aligning with the growing global demand for digital services. Mangalore's emergence as a cost-effective IT hub, benefiting from a skilled local talent pool and government support, provides a favorable economic environment for Technomers to leverage its technical expertise and innovation focus for potential growth within this dynamic regional market.

Porter's Five Forces Framework

Porter's Five Forces is a business framework that helps entrepreneurs shape their strategy to drive profitability. The framework is a holistic way of looking at any industry and understanding the structural underlying drivers of profitability and competition.

The **5 porters** analysis of the IT industry is as follows:

1. Competitive Rivalry:

This force examines the intensity of competition among existing players in the industry. Factors like the number of competitors, their relative size and market share, and the extent of their product differentiation impact this force.

2. Threat of New Entrants:

This force assesses the ease with which new companies can enter the market. Factors like the cost of entry, regulatory barriers, and the existence of strong brand loyalty among customers play a role.

3. Bargaining Power of Suppliers:

This force considers the influence that suppliers have on the industry. Factors like the number of suppliers, the availability of substitutes, and the cost of switching suppliers impact their power.

4. Bargaining Power of Buyers:

This force examines the influence that buyers have on the industry. Factors like the number of buyers, the price sensitivity of buyers, and the availability of substitutes for the product or service impact their power.

5. Threat of Substitute Products or Services:

This force evaluates the likelihood of customers switching to alternative products or services that fulfill the same need. Factors like the availability of substitutes, the price and quality of substitutes, and the ease of switching to substitutes impact this force.

To name a few strong 5 forces of porters analysis for IT industry are:

- High innovation rate leads to a number of competitors i.e Threat to new entrants and Industry rivalry
- There will be a different version of the emergent software/tech or a new software or tech i.e Threat of substitute
- As there are multiple vendors with similar products it leased to increased buyer option i.e Bargaining power of the buyer
- Shortage of talent gives power to skilled workers to negotiate i.e Bargaining power of supplier



Fig 1.2 Porters 5 Forces

1.1.3 Competitor Analysis

A competitor analysis of a company would involve identifying direct and indirect competitors, examining their strengths and weaknesses, and understanding their market positioning. This analysis helps a company understand its own competitive landscape and identify opportunities to gain a market edge.

Technomers in mangalore is a provider of cloud, AI, DevOps, and other services. Some of the top competitors, each offering services in the same area are as follows:

1. Infosys

- **Strengths:** A global leader in IT services with a significant presence in Mangalore. Offers a comprehensive suite of services including cloud solutions (Infosys Cobalt), AI and machine learning, cybersecurity, and enterprise Agile DevOps.
- **Notable Offerings:** Application development, cloud infrastructure, data analytics, and digital transformation services.

2. Cognizant Technology Solutions

- **Strengths:** A multinational IT services company with a robust portfolio in cloud computing, AI, software engineering, and automation. The Mangalore branch contributes significantly to its Indian operations.
- **Notable Offerings:** Cloud computing, AI, IT infrastructure management, and automation services.

3. Mphasis

- **Strengths:** Specializes in application services, AI, modernization, and cloud solutions. Known for delivering sustainable software and technology solutions across various industries.
- **Notable Offerings:** Application services, AI, DevOps, cloud services (AWS, Azure, GCP), and enterprise automation.

4. Novigo Solutions

- **Strengths:** A fast-growing IT services company known for delivering digital transformation solutions, with expertise in AI, application development, and robotic process automation.
- **Notable Offerings:** AI and machine learning, application development, cloud services, and IT consulting.

5. Raasoft Infotech Pvt Ltd

- **Strengths:** Focuses on custom software development, mobile and web development, IT consulting, and system integration. Utilizes modern technologies like Ruby on Rails, Python, and AI.
- **Notable Offerings:** Web and enterprise application development, cloud application integration, fintech solutions, and business process automation.
-

1.2 Company Analysis

Technomers is an emerging IT services company based in Mangalore, offering solutions in cloud computing, artificial intelligence (AI), DevOps, and digital transformation. The following analysis outlines the company's internal strengths and weaknesses, as well as the external opportunities and threats it faces in the competitive landscape. Technomers has not raised any funding yet. The company has 88520 active competitors, including 888 funded and 5393 that have exited.

SWOT Analysis of Technomers

Strengths

- **Diverse Service Portfolio:** Technomers provides a comprehensive range of services including cloud infrastructure, AI, DevOps, and digital automation, catering to the evolving needs of both startups and enterprises.
- **Agile and Adaptable:** Being a relatively young and nimble company, Technomers can quickly adopt emerging technologies and pivot their strategies to stay relevant in a rapidly changing tech environment.
- **Local Market Understanding:** With its roots in Mangalore, Technomers has a strong grasp of regional business dynamics and a growing network of local clients.

Weaknesses

- **Limited Brand Recognition:** Compared to established IT giants in the region, Technomers still lack widespread brand awareness and global visibility.
- **Resource Constraints:** As a growing firm, it may face limitations in terms of staffing, infrastructure, and access to large-scale enterprise clients.
- **Client Dependence:** A small or concentrated client base could make the company vulnerable to revenue fluctuations if key accounts are lost.

Opportunities

- **Growing Demand for Cloud and AI:** The global shift towards cloud-native and AI-driven solutions opens up vast growth opportunities across multiple industries.
- **Digital Transformation in SMEs:** Small and medium enterprises in India and abroad are increasingly seeking affordable and effective digital solutions, an area where Technomers can excel.
- **Government Incentives:** Supportive policies under India's Digital India and Startup India

initiatives offer potential funding, mentorship, and exposure.

Threats

- **Intense Competition:** The presence of large IT firms in Mangalore and Bangalore creates stiff competition in terms of talent acquisition and client retention.
- **Rapid Technological Changes:** Staying updated with the fast pace of innovation in AI, cybersecurity, and DevOps requires continuous investment and training.
- **Cybersecurity Risks:** As a provider of cloud and digital services, Technomers is inherently exposed to cybersecurity threats that could impact its reputation and client trust.

PESTEL Analysis of Technomers

This framework evaluates the macro-environmental factors that can impact Technomers' strategic direction.

Political

- The Indian government's emphasis on digital infrastructure and startup ecosystems has created a favorable environment for tech firms.
- Policy stability in Karnataka, combined with state-level IT incentives, supports local firms like Technomers.
- However, any changes in international trade relations or data localization laws may affect operations with overseas clients.

Economic

- India's growing digital economy and increased IT spending by both the public and private sectors provide a strong foundation for Technomers' growth.
- Rising inflation and wage pressures in the tech industry could impact profitability.
- The demand for cost-effective IT solutions during global economic slowdowns can work in favor of smaller, agile providers like Technomers.

Social

- Increased digital literacy and demand for tech solutions across sectors, including education, healthcare, and logistics, are expanding Technomers' potential market.
- A growing young talent pool in Mangalore and surrounding regions offers recruitment

advantages.

- Shifts in client expectations toward seamless and personalized digital experiences require continuous innovation and customer-centricity.

Technological

- Rapid advancements in AI, cloud computing, and DevOps create both opportunities and pressure to innovate constantly.
- Open-source platforms and automation tools reduce development time but increase competition.
- Investment in R&D and collaboration with academia or tech incubators can be strategic levers for innovation.

Environmental

- As sustainability becomes a key concern, clients are demanding greener, energy-efficient IT practices.
- Technomers can gain a competitive edge by offering low-energy cloud solutions and promoting green IT policies.
- Environmental regulations related to data centers and electronic waste disposal may impact operations in the future.

Legal

- Compliance with data protection laws (e.g., India's Digital Personal Data Protection Act, GDPR for EU clients) is essential.
- IP protection, cybersecurity standards, and software licensing laws are critical areas for risk management.
- Government audits and procurement policies can affect the bidding and execution of public-sector contracts.

CHAPTER 2

CONCEPTUAL BACKGROUND

2.1 Systematic literature review

A **systematic literature review (SLR)** methodology involves a structured, transparent, and replicable approach to identifying, selecting, and critically analyzing research studies to answer a specific research question. It uses a predefined protocol, including search strategies, inclusion/exclusion criteria, and data extraction methods, to ensure a comprehensive and unbiased review of the literature.

The systematic literature review follows PRISMA framework. PRISMA provides a standard methodology that contributes to the quality assurance of the revision and replication process. A systematic review was developed by explaining the article selection criteria, search strategy, data extraction, and data analysis procedures. In general, the PRISMA method is divided into several steps as follows:

1. Determine the source of information
2. Study selection
3. Data collection process
4. Determine eligibility criteria
5. Selection of data items

Bibliometric analysis is a quantitative method used to evaluate and interpret patterns within academic literature. In this project, bibliometric analysis was conducted to explore the research landscape around machine learning applications in wearable cardiac monitoring. Using data sourced from the Scopus database, the analysis helped identify key trends, influential authors, frequently used keywords, and popular machine learning algorithms. VOSviewer, a specialized tool for visualizing bibliometric data, was used to generate keyword co-occurrence maps, author collaboration networks, and thematic clusters. This analysis provided valuable insights into which technologies and approaches are most prominent in the field, helping guide the selection of suitable

models like Random Forest and XGBoost for further implementation. Overall, bibliometric analysis served as an evidence-based foundation for decision-making and model selection in the internship project.

Our research focused on getting solution for cardiovascular diseases(CVD) by monitoring the heart health using wearable devices/technology using machine learning. It was found from our analysis that most popularly used technique for analysing heart data to determine disease is random forest technique, Xgboost technique and SVM (support vector machine).

2.2 Machine learning

Machine Learning, often abbreviated as ML, is a subset of artificial intelligence (AI) that focuses on the development of computer algorithms that improve automatically through experience and by the use of data. In simpler terms, machine learning enables computers to learn from data and make decisions or predictions without being explicitly programmed to do so. Unlike conventional machine learning algorithms, deep learning algorithms are less linear, more complex, and hierarchical, capable of learning from enormous amounts of data, and able to produce highly accurate results.

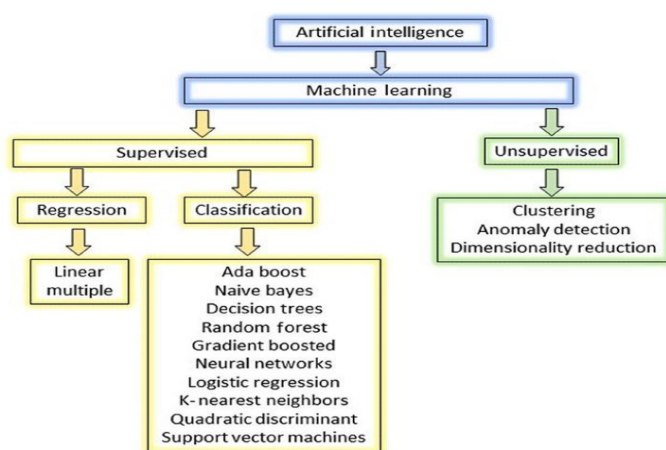


Fig 2.1 Artificial Intelligence

Random forest:

Random Forest Classifier

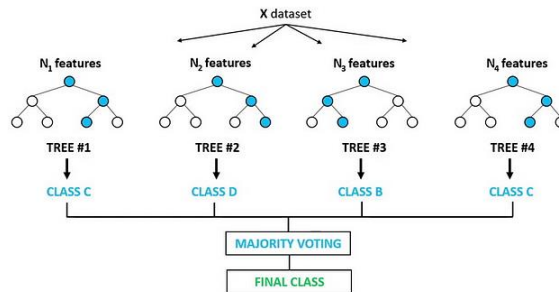


Fig 2.2 Random forest classifier

A popular machine learning technique for both classification and regression, random forest is strong and adaptable. The name comes from the way it aggregates the predictions of several decision trees to create a “forest” of them. A random selection of characteristics and data points are used to train each tree in the forest, which increases overall prediction robustness and reduces overfitting when compared to a single decision tree.

XGBoost:

XGBoost is an ensemble learning algorithm that combines the predictions of multiple weak predictive models, called base learners, to create a strong predictive model. It belongs to the class of boosting algorithms, which iteratively trains weak models and assigns higher weights to misclassified instances to improve subsequent models.

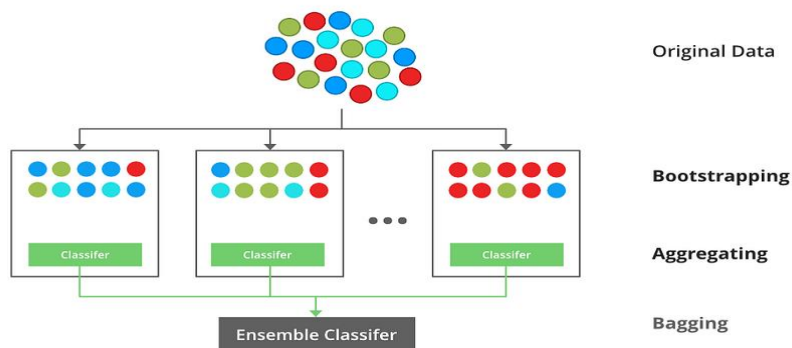


Fig 2.3 XGboost

Support Vector Machine:

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. While it can handle regression problems, SVM is particularly well-suited for classification tasks. SVM aims to find the optimal hyperplane in an N-dimensional space to separate data points into different classes. The algorithm maximizes the margin between the closest points of different classes.

2.3 Software Tools

The data required for the exploratory analysis is collected from **UCI Machine Learning Repository** is a comprehensive online collection of datasets used for machine learning research, development, and education. It is a collection of databases, domain theories, and data generators used by the machine learning community for empirical analysis of machine learning algorithms. The archive was created as an ftp archive in 1987 by UCI PhD student David Aha. Since that time, it has been widely used by students, educators, and researchers all over the world as a primary source of machine learning datasets. It's a free and publicly accessible resource, offering a wide variety of datasets for various machine learning tasks, making it a valuable tool for researchers, students, and practitioners in the field.

VOSviewer is a software tool used to visualize and analyze bibliometric data, such as keyword co-occurrence, author networks, and research trends. It creates visual maps that help identify patterns and relationships within large sets of scientific literature. In this project, VOSviewer was used to analyze Scopus data and generate insights through keyword and author collaboration mapping.

Visual Studio is an Integrated Development Environment (IDE) developed by Microsoft to develop Desktop applications, GUI (Graphical User Interface), console, web applications, mobile applications, cloud, and web services, etc. With the help of this IDE, you can create managed code as well as native code. It uses the various platforms of Microsoft software development software

like Windows store, Microsoft Silverlight, and Windows API, etc. It is not a language-specific IDE as you can use this to write code in C#, C++, VB (Visual Basic), Python, JavaScript, and many more languages. It provides support for 36 different programming languages. It is available for Windows as well as for macOS.

Scopus is a commercial database for peer-reviewed literature produced by Elsevier. It covers articles till 2024 and fully covers Frontier, EMBASE, MDPI ScienceDirect and various such citation database from various sources, including journals, books, and conference proceedings. It covers a wide range of subjects, making it a versatile resource for various research areas at the same time also focusing on the quality of the content.

CHAPTER 3

PROJECT DESIGN & METHODOLOGY

3.1 PROJECT OBJECTIVES

1. Investigating the role of machine learning in the early detection and monitoring of cardiovascular diseases (CVD) using physiological data collected from wearable devices along with data-driven approaches for proactive healthcare.
2. To conduct a systematic literature review to identify the most commonly used and effective machine learning algorithms for CVD detection, such as Random Forest, XGBoost, and Support Vector Machines and evaluating based on their performance.
3. Conducting exploratory data analysis using the WESAD dataset, to understand how bio-signals vary across different emotional and physical states (e.g., rest, stress, amusement) and how machine learning models can classify these states. Through model training, testing, and performance evaluation.
4. To demonstrate how machine learning, combined with wearable technology, can support early detection of cardiovascular conditions, highlighting the current challenges in data quality, feature engineering, model generalizability, and real-world deployment, offering suggestions for future improvements in personalized digital health solutions.

This project aims to bridge the gap between raw physiological signal data and actionable health insights by reviewing existing literature and analyzing real-world datasets to understand the effectiveness of machine learning techniques in this domain. the project seeks to identify key features that contribute to accurate classification and understand the limitations of using basic statistical features.

3.2 METHODOLOGY

The study began with a review of Scopus studies examining published research that focused on the

collection of heart disease data and heart data using wearables in children and/or adolescents/adults. The search strategy included the following terms: ("wearable?" OR "devic*" OR "technolog*") AND ("machine learning" OR "AI" OR "decision tree" OR "random forest" OR "gradient boosting" OR "XGB" OR "support vector machine" OR "artificial intelligence" OR "big data analytics" OR "data mining") AND ("heart" OR "cardiac")

Systematic literature review Process

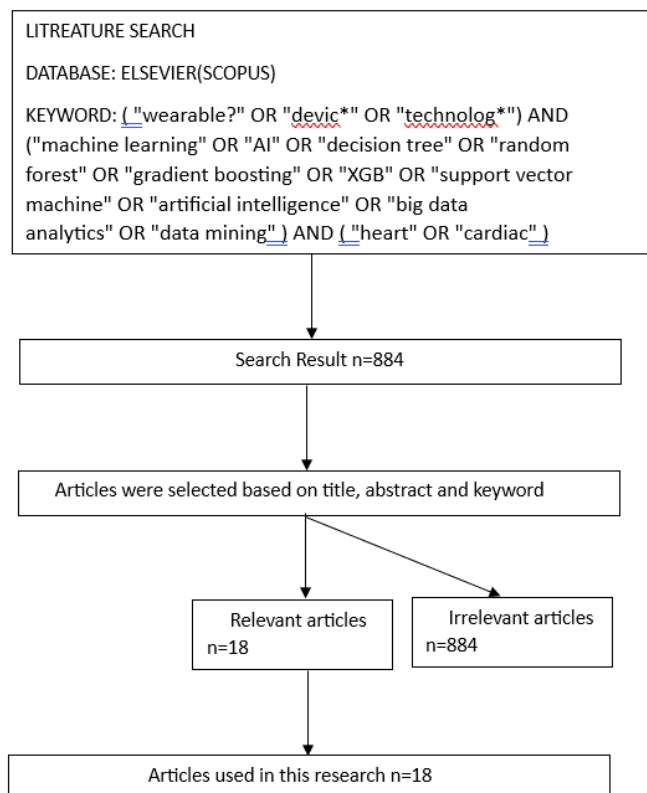


Fig 3.1 Systematic literature review procedure

The search produced 884 articles (2015- 2024), of which 71 articles were included in the analysis, looking further into them 18 articles completely matched our research idea and 53 articles were rejected. The majority of the evidence involved the use of wearable technology using machine learning to detect heart disease. The usage of machine learning in wearables helped in capturing a variety of data, for monitoring, detecting, predicting, and preventing cardiovascular events.

A bibliometric analysis of the Scopus data was also performed using VOSviewer software to

generate various analytical reports, including keyword analysis, author analysis, and abstract term analysis. To facilitate this analysis, a custom Python script was developed to convert the Excel data—containing titles, authors, author keywords, and abstracts—into a CSV (comma-separated values) file. This process generated several output files: *vosviewer_abstract_terms*, *vosviewer_authors*, *vosviewer_keywords*, and *vosviewer_terms*. These text files were then imported into the VOSviewer software to produce keyword analysis, related keyword analysis, author analysis, co-author analysis, keyword co-occurrence analysis, and author co-occurrence analysis.

To gain hands-on experience with exploratory analysis, the WESAD dataset relevant to the topic selected for the systematic literature review was collected from the UCI Machine Learning Repository. Python programming was used within Visual Studio to conduct the exploratory analysis, focusing on Respiration Analysis and Recurrence Analysis of respiratory data obtained using the RespiBan device.

Table 3.1 Steps of RespiBAN dataset for the exploratory analysis

Step	What we did	Why we did
1. Loaded RespiBAN data	Read S2_respiban.txt carefully (ignoring # comments)	Got access to raw sensor signals
2. Identified real respiration signal	Auto-detected the correct column based on slow variation (few zero crossings)	Found the true respiration waveform
3. Preprocessed respiration	Centered (removed mean), downsampled (700 Hz → 35 Hz), bandpass filtered (0.05–0.8 Hz)	Cleaned the breathing signal for clear cycles
4. Respiratory pattern analysis	Plotted breathing waveform	Visually verified correct breathing oscillations
5. Generated Recurrence Plot (RP)	Created a binary matrix showing when signal returned to similar states	Visualized breathing rhythm stability
6. Extracted Recurrence Metrics (RR, DET)	Calculated Recurrence Rate and Determinism	Quantified how repetitive and predictable breathing was

Recurrence Plot is a **visual tool** used to analyze **repeating patterns** in time-series data — like respiration, heart rate, or brain activity.

Method to Read a Recurrence Plot

1. Axes = Time

- Both X and Y axes represent **time (sample index)**.
- Each point (i, j) in the matrix indicates whether the signal at time i is **similar** to the signal at time j.

2. Black Dots = Recurrence

- A **black dot** at (i, j) means the signal at those two time points is similar — i.e., the signal **returns to a previous state**.
- The **density and arrangement** of these dots reveal important patterns.

3. Diagonal Lines = Repeating Patterns

- **Long diagonal lines** show **predictable and periodic** behavior.
 - Example: **regular breathing cycles**
- **Short or broken diagonals** mean **less regularity** or **transitions**.

4. Vertical or Horizontal Lines = Pauses

- These may indicate **pauses, plateaus, or slow-changing states** (e.g., breath holding or slow inspiration).

5. Grid-Like Patterns = Strong Periodicity

- Your RP shows **strong, symmetric, grid-like patterns**, especially in the center.
 - This reflects **stable, self-similar respiration** over that period.
 - Likely from a **neutral or calm state**.

6. Transitions at Edges or Blocks

- The **top-right and bottom-left corners** have **denser or different recurrence structures**.

- These could represent **transitions** between states (e.g., moving from neutral → stress → amusement).
- Or simply a **change in breathing behavior** over time.

The basic steps carried in the code to generate the output for **Personalized Activity Type Detection** is as follows:

1. Data Collection

Two datasets were used:

- **HR.csv** — Heart rate data sampled over time.
- **EDA.csv** — Electrodermal activity (skin conductance), which reflects arousal levels (e.g., stress, excitement).

2. Preprocessing & Feature Extraction

- Both signals were resampled into **30-second time windows**.
- For each window, the following statistical features were extracted:
 - Mean and standard deviation of HR
 - Mean and standard deviation of EDA
- This resulted in a structured feature set representing physiological activity in each time window.

3. Labeling (Simulated for Demonstration)

Since ground-truth activity labels were not available, simulated labels were created to represent three typical activity states:

- **0** → Resting
- **1** → Mentally active (e.g., focused tasks)
- **2** → Physically active (e.g., walking or movement)

These simulated labels allowed us to demonstrate the feasibility of the classification approach.

4. Model Training

Random Forest classifier, an XGBoost model, and a Support Vector Machine were trained on the extracted features to learn the patterns corresponding to each activity type.

5. Evaluation

- The model's predictions were tested on unseen data.
- The model was evaluated based on accuracy, confusion matrix, and a feature importance

chart.

- The results demonstrated that physiological signals alone were sufficient to classify activity type with high accuracy.

Insights from the systematic literature review indicated that the majority of cardiac disease detection studies achieved high accuracy using machine learning methods, particularly the Random Forest algorithm. Consequently, the analysis conducted using the WESAD dataset primarily focused on the Random Forest method and XGBoost for personalized activity type detection. Classification reports for each model were generated to facilitate comparison.

Descriptions of key terminologies in Classification Report Metric :

- **Accuracy**
Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted instances to the total number of predictions. While useful, it can be misleading in imbalanced datasets.
- **Macro Average Precision**
This is the average precision across all classes, treating each class equally. It doesn't consider class imbalance, so it gives equal weight to all classes regardless of their support.
- **Macro Average Recall**
This calculates the average recall for all classes, giving a balanced view of the model's ability to correctly identify each class, regardless of how many samples each class has.
- **Macro Average F1-Score**
The macro average F1-score is the mean of the F1-scores for all classes. It provides a single measure that balances precision and recall equally across all classes.
- **Weighted Average Precision**
This is the average precision for all classes, weighted by the number of true instances in each class (support). It gives a more realistic picture in imbalanced datasets.
- **Weighted Average Recall**

Similar to macro recall, but gives more importance to classes with more samples. It tells how well the model is identifying true positives across the dataset.

- **Weighted Average F1-Score**

This metric combines weighted precision and recall to give a balanced measure of model performance, considering both accuracy and class distribution.

The goal of this analysis was to use physiological signals—specifically **heart rate (HR)** and **electrodermal activity (EDA)**—to detect and classify different types of user activity (such as resting, mental work, or physical activity) using machine learning. This analysis helps demonstrate how bio-signals can be interpreted to understand behavior without needing wearable motion sensors or manual labeling.

This has practical use in:

- **Mental health monitoring**
- **Fatigue or productivity tracking**
- **Context-aware systems and smart devices**
- **Non-invasive behavior analysis**

3.3 WORK DONE

Initially there was a foundational training was in Python, SQL, Seaborn, and Matplotlib which provided me with the essential analytical tools leveraged by the firm. These courses were completed on coursera post which for the systematic literature review, we conducted a bibliometric analysis of the scopus data collected.

Why bibliometric analysis in systematic literature review:

Researchers use bibliometric analysis to:

- Determine the current state of a field of study.
- Identify influential publications, authors, and institutions.

- Analyze research trends and identify emerging areas of research.
- Evaluate the impact of research and publications.
- Understand the structure and relationships within a field of research.

- **Data Sources:**

Bibliometric analysis relies on data from various databases, such as:

- Web of Science.
- Scopus.
- PubMed.
- Google Scholar.
- Dimensions.

The data source used in this project is scopus for its easy access to open source articles.

- **Software Tools:**

Several software tools can be used to conduct bibliometric analysis, including:

- VOSviewer.
- CiteSpace.
- BibExcel.
- HistCite.
- Publish or Perish.

The bibliometric analysis tool used is VOSviewer.

Why use Machine learning techniques for data analysis of wearable device ?

The data collected is crucial for understanding health trends, personalizing treatments, and improving healthcare outcomes through insights into individual health patterns, activity levels, and physiological metrics. The estimated data analysis to be conducted is as follows

A. Wearable Device Data Analysis

- **Data Cleaning & Preprocessing:** Handle missing values, normalize sensor data, and align timestamps.
- **Descriptive Statistics:** Summary of patient metrics (EDA, temperature, PPG, accelerometer, heartbeat patterns).
- **Time-Series Analysis:** Identify trends, anomalies, and correlations between different physiological signals.
- **Predictive Modeling:** Develop machine learning models to predict stress, abnormal heart conditions, or activity levels.
- **Clustering/Segmentation:** Group patients based on physiological patterns to find meaningful clusters (e.g., high-risk vs. low-risk individuals).

Using Vosviewer the bibliometric analysis was done. It helped us to get a visualisation and also get bibliometric insights from the scopus data collected. To conduct the bibliometric analysis of the Scopus data a Python code was written to convert the Excel data with title, author, author keyword, and abstract into a CSV(comma-separated value) file. This file was loaded into the Vosviewer software to generate the keyword analysis, related keyword analysis, author analysis, co-author analysis, keyword co-occurrence analysis, and author co-occurrence analysis.

3.2.1 CHARTS

Charts from the articles selected for bibliometric analysis

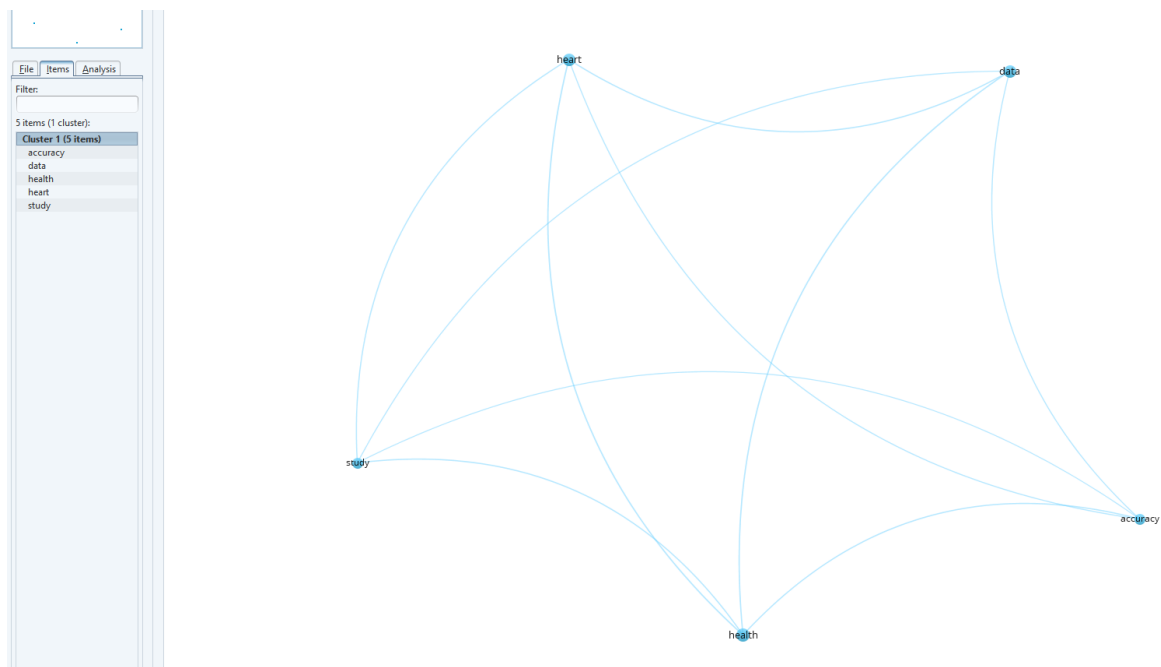


Fig 3.2 Abstract terms analysis

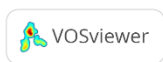
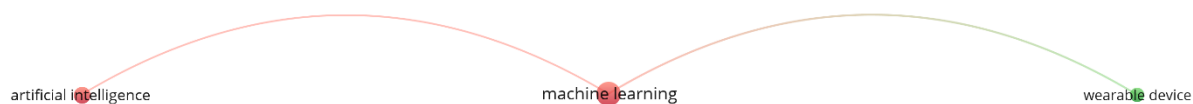


Fig 3.3 Keyword analysis

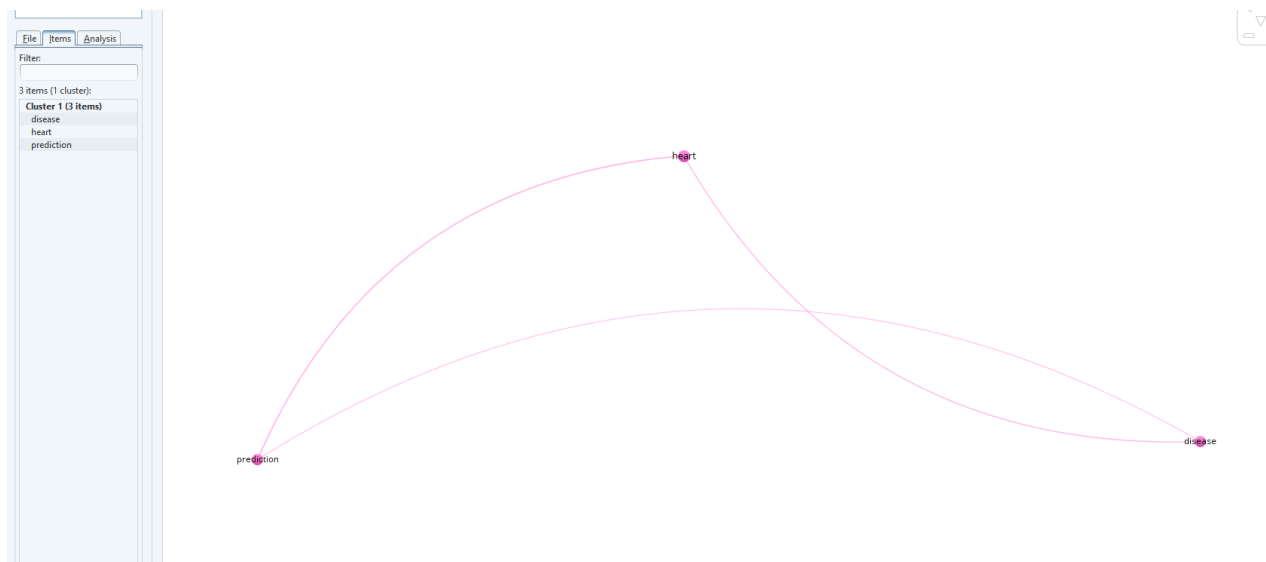


Fig 3.4 Title_term_analysis

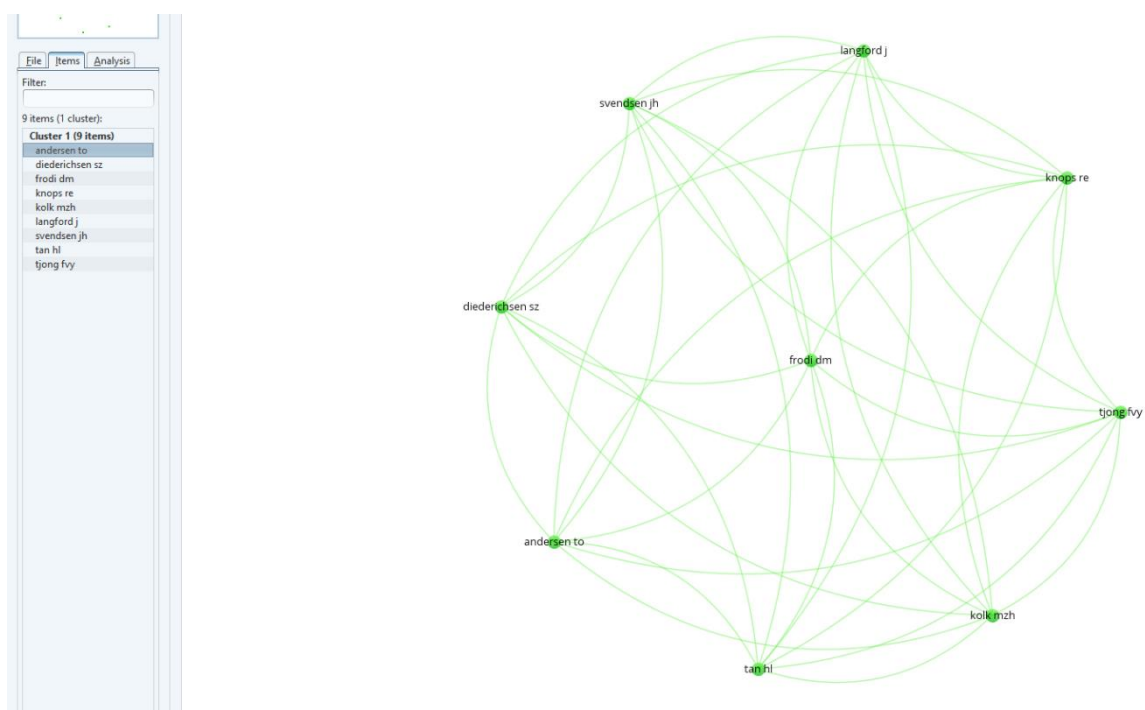


Fig 3.5 Author_analysis

3.2.2 Charts from the entire Scopus file generated for bibliometric analysis

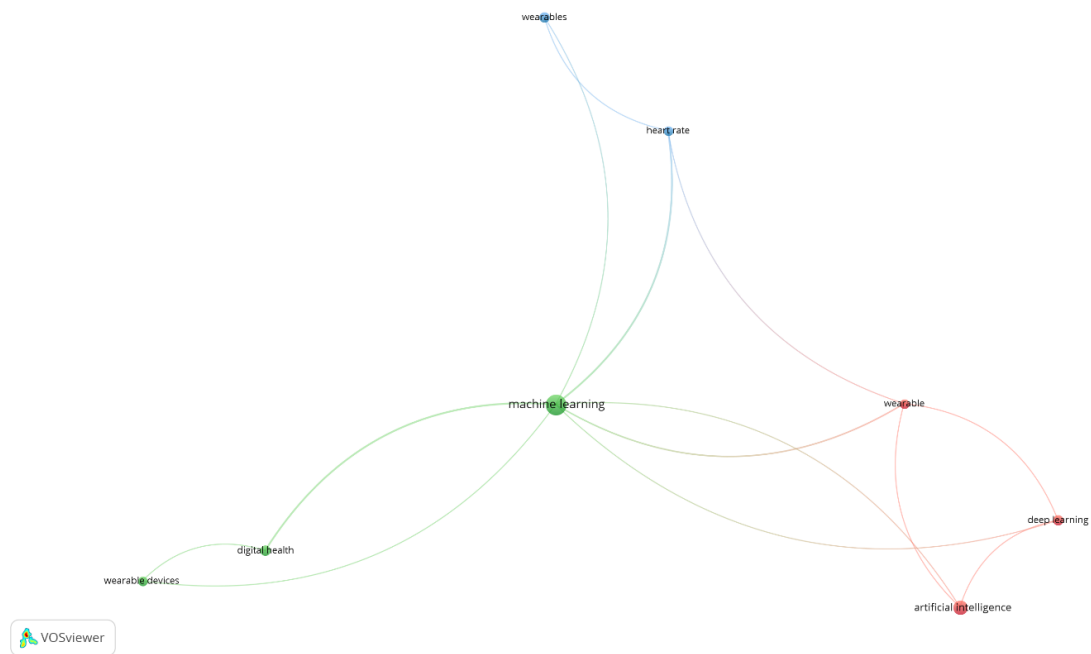


Fig 3.6 Co occurrence author keyword analysis

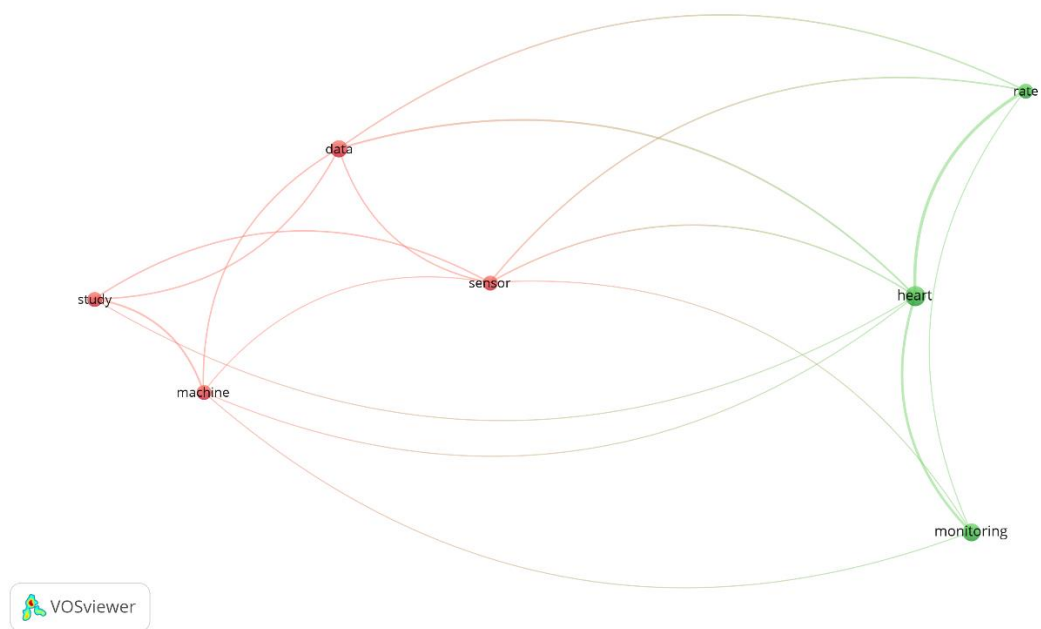


Fig 3.7 Article terms analysis

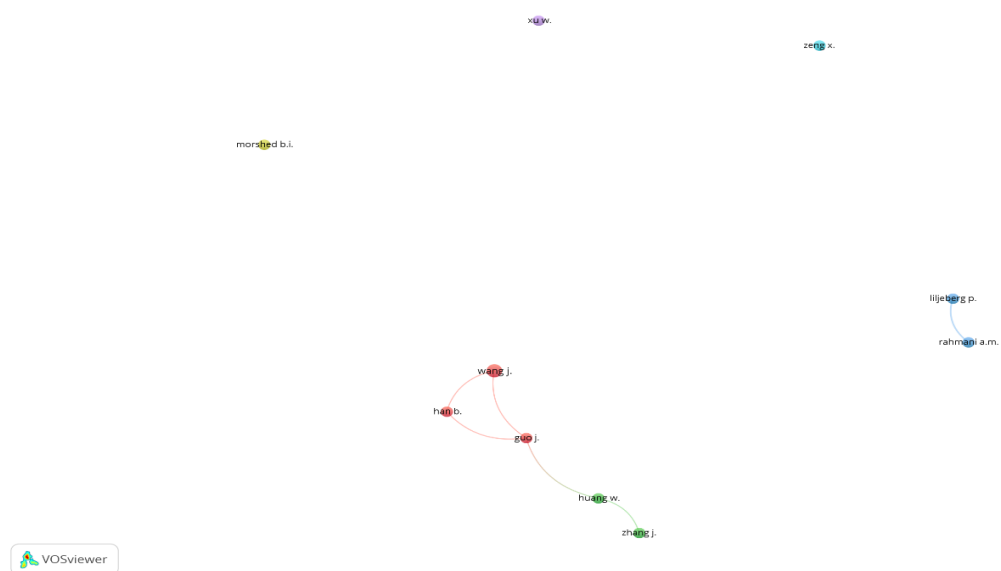


Fig 3.8 Author analysis

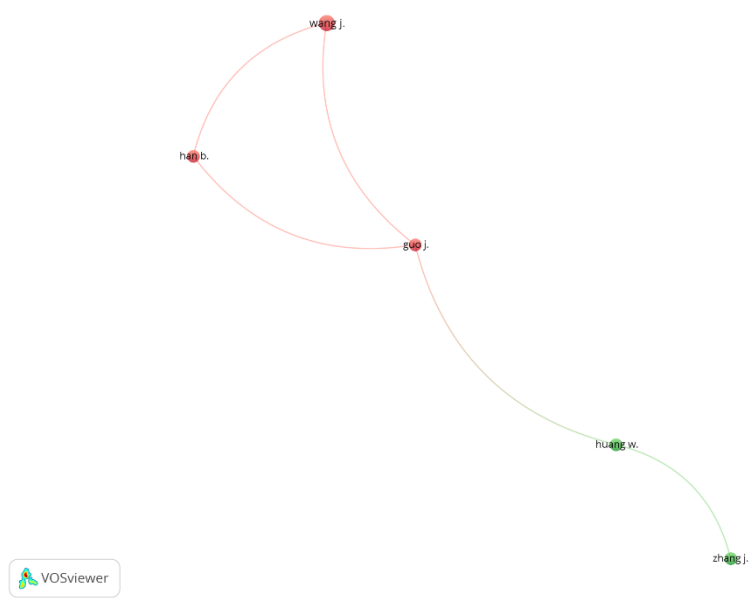


Fig 3.8.1 Related Author



Fig 3.8.2 Unrelated Author

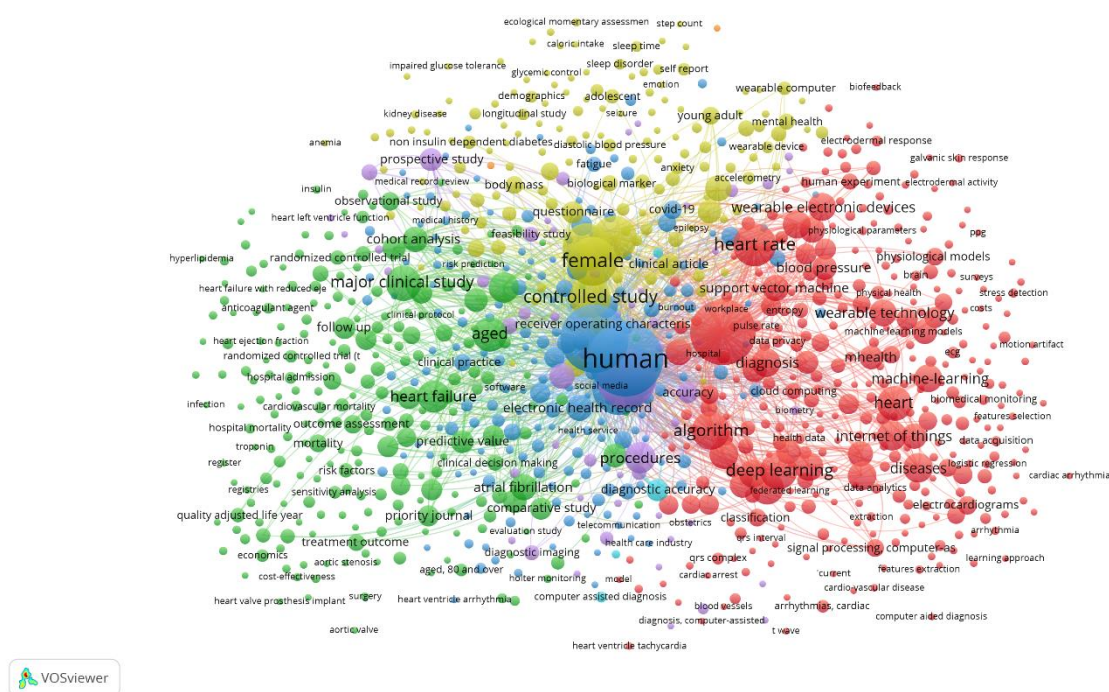


Fig 3.9 Complete Scopus search analysis

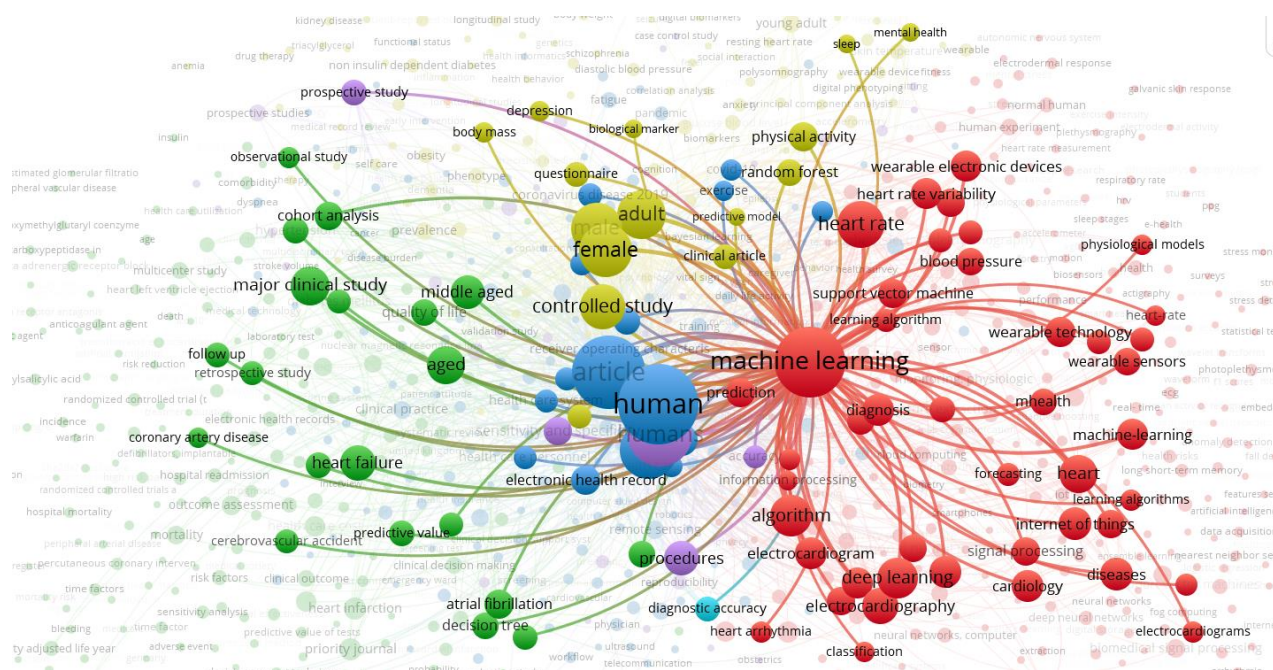


Fig 3.10 Scopus machine learning related search

Description of the bibliometric analysis insight used:

Common methods used in bibliometric analysis include:

- **Citation analysis:** Analyzing how often publications are cited to determine their impact.
- **Co-citation analysis:** Identifying groups of publications that are frequently cited together, suggesting a shared focus or influence.
- **Bibliographic coupling:** Examining publications that cite the same sources, revealing relationships between them.
- **Keyword analysis:** Analyzing the keywords used in publications to identify key themes and topics.
- **Co-authorship analysis:** Examining collaboration patterns among researchers.
- **Abstract term analysis:** Analysis of the most widely used terms in the abstract.
- **Title term analysis:** Analysis of the most widely used terms in the title.
- **Article term analysis:** Analysis of the most widely used terms in the article.

Python programming was used to generate insights from respiratory analysis, recurrence plot analysis, and Random Forest analysis on data from the RespiBan and Empatica E4 devices, which was obtained from the UCI Machine Learning Repository. The corresponding Python code is provided below.

3.2.3 INDEX FIGURE

Below is the **exploratory analysis** python code for a Patient 2 from WESAD dataset
Result is there in the **findings** section

→Respiban

```

##: Load and Preprocess RESP Signal ##
import pandas as pd
from scipy.signal import butter, filtfilt
from scipy.signal import butter, filtfilt
import matplotlib.pyplot as plt
import numpy as np

# Load data
df = pd.read_csv("C:/Users/gunja/Desktop/Tecnomers internship/systematic lit review/WESAD DATA analysis/S5/S5_respiban.txt", delimiter="\t", comment="#", header=None)

resp_raw = df[9].values
resp_raw = resp_raw - np.mean(resp_raw) # Center the signal

# Step 2: Check the number of columns
print(f"Number of columns detected: {df.shape[1]}")

# Step 3: Print first few rows
print(df.head())

# Step 4: Find which columns have valid (non-NaN) data
#valid_columns = df.columns[df.notna().any()].tolist()
valid_columns = df.columns[2:10].tolist()
print(f"Valid columns with real data: {valid_columns}")

# Step 5: Plot each valid column quickly to visually identify respiration
for col in valid_columns:
    plt.figure(figsize=(10, 3))
    plt.plot(df[col].values[:3000]) # Plot first 3000 samples
    plt.title(f"Quick Plot - Column {col}")
    plt.xlabel("Sample Index")
    plt.ylabel("Signal Value")
    plt.grid(True)
    plt.show()

```

Fig 3.11 Code Outputting graph of 9 individual graph parameter and column value

Table 3.2 Output

Number of columns detected: 11

Row	Col_0	Col_1	Col_2	Col_3	Col_4	Col_5	Col_6	Col_7	Col_8	Col_9	Col_10
0	0	0	31053	14694	32651	29553	37415	32324	31659	31364	NaN
1	1	0	30861	14680	32983	29556	37417	32333	31663	31379	NaN
2	2	0	30644	14684	32823	29561	37409	32340	31655	31367	NaN
3	3	0	30374	14701	32569	29553	37405	32341	31671	31076	NaN
4	4	0	30103	14684	32794	29561	37409	32342	31681	31365	NaN

Valid columns with real data: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

In the above program the WESAD respiban data of patient 2 was loaded using python programming. The file command was used to read the file, and then the data is printed the column 0-10 for 5 patients. The column 3,4,5,6,7,8,9,10 are the columns of the data collected using the respiban device from the patients using the "sensor": ["ECG", "EDA", "EMG", "TEMP", "XYZ", "XYZ", "XYZ", "RESPIRATION"] which correspond to the channel here as follows "channels": [1, 2, 3, 4, 5, 6, 7, 8]. This code also outputs the individual graphs for the parameters ["ECG", "EDA", "EMG", "TEMP", "XYZ", "XYZ", "XYZ", "RESPIRATION"] in the respiban file, which is displayed in the findings section.

```

# Load data
df = pd.read_csv("C:/Users/gunja/Desktop/Tecnomers internship/systematic lit review/WESAD DATA analysis/S2/S2_respiban.txt", delimiter="\t", comment="#", header=None)

# (Manual) Assume respiration is in column 8
resp_raw = df[9].values
resp_raw = resp_raw - np.mean(resp_raw) # Center the signal

# Downsample
fs = 700
downsample_factor = 20
fs_ds = fs // downsample_factor
resp_ds = resp_raw[::downsample_factor]

# Bandpass filter
def bandpass_filter(signal, lowcut, highcut, fs, order=4):
    nyq = 0.5 * fs
    low = lowcut / nyq
    high = highcut / nyq
    b, a = butter(order, [low, high], btype="band")
    return filtfilt(b, a, signal)

resp_filtered = bandpass_filter(resp_ds, 0.05, 0.8, fs_ds)

# Plot
plt.figure(figsize=(12, 4))
plt.plot(resp_filtered[:fs_ds*60])
plt.title("Filtered Respiration Signal (First 60 seconds)")
plt.xlabel("Sample Index")
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()

```

Fig 3.12 Code Outputting graph of respiration signal analysis

The above code outputs the respiration signal analysis graph. The respiratory data in the respiban data was sampled at 700Hz which was very high and it wastes memory and computation. Hence it was downsampled to 35Hz by dividing the respiratory column by 20 which is stored in the fs_ds variable. The bandpass filtering is done to clean the signal data and remove the noise for smooth feature extraction.

```

from pyts.image import RecurrencePlot

rp = RecurrencePlot(threshold='point', percentage=20)
resp_resaped = resp_filtered[:35*60].reshape(1, -1) # 60s window
X_rp = rp.fit_transform(resp_resaped)

plt.imshow(X_rp[0], cmap='binary', origin='lower')
plt.title("Recurrence Plot of Respiration")
plt.show()

#Extract Breathing Features (Use scipy.signal.find_peaks)
#You can extract breath rate, amplitude, and variability.
#Python Code: Breathing Feature Extraction
from scipy.signal import find_peaks
# Use the filtered respiration signal from before
signal = resp_filtered

# Detect peaks (inhalation points)
peaks, _ = find_peaks(signal, distance=fs_ds*1.5) # 1.5s min between breaths

# Calculate breath-to-breath intervals (in seconds)
bbi = np.diff(peaks) / fs_ds

# Compute breathing rate (breaths per minute)
breathing_rate_bpm = 60 / np.mean(bbi) if len(bbi) > 0 else np.nan

# Compute amplitude of each breath
amplitudes = signal[peaks] - np.min(signal) # relative to baseline

```

```

# Feature summary
features = {
    "breathing_rate_bpm": round(breathing_rate_bpm, 2),
    "mean_amplitude": round(np.mean(amplitudes), 2),
    "amplitude_std": round(np.std(amplitudes), 2),
    "mean_bbi_sec": round(np.mean(bbi), 2),
    "bbi_std": round(np.std(bbi), 2),
    "num_breaths_detected": len(peaks)
}

# Display
for k, v in features.items():
    print(f"{k}: {v}")

```

Fig 3.13 Recurrence plot of complete respiratory signal

The above code takes the filtered respiratory data from the respiratory signal analysis code gives an output for recurrence plot of the respiration signal along with the summarized feature values calculated for the recurrence plot. The explanation of the same is given in the findings section.

```

#Comparison of Breathing Features by Segment
#Step 2: Define Segment Windows
# Each segment = 5 minutes = 300 seconds + 300 * 35 samples
segment_length = 300 * fs_ds # fs_ds = 35 Hz
segments = {
    "neutral": resp_filtered[0 : segment_length],
    "amusement": resp_filtered[segment_length : 2*segment_length],
    "stress": resp_filtered[2*segment_length : 3*segment_length]
}

#Step 3: Extract Features for Each Segment
def extract_breathing_features(signal, fs):
    peaks, _ = find_peaks(signal, distance=fs*1.5)
    bbi = np.diff(peaks) / fs if len(peaks) > 1 else [0]
    amplitudes = signal[peaks] - np.min(signal)
    return {
        "breathing_rate_bpm": round(60 / np.mean(bbi), 2) if np.mean(bbi) > 0 else np.nan,
        "mean_amplitude": round(np.mean(amplitudes), 2),
        "amplitude_std": round(np.std(amplitudes), 2),
        "mean_bbi_sec": round(np.mean(bbi), 2),
        "bbi_std": round(np.std(bbi), 2),
        "num_breaths": len(peaks)
    }

# Apply to all segments
features_by_segment = {label: extract_breathing_features(sig, fs_ds) for label, sig in segments.items()}

# Display
#import pandas as pd
df_features = pd.DataFrame(features_by_segment).T
print(df_features)

```

Fig 3.14 Comparison of Breathing Features by Segment

The above code gives an output for the breathing feature values like breathing rate, amplitude, number of breath, breath-to-breath-interval and explanation of the same is given in the findings section.

Analysis done using data collected of patient no.4, the data is collected using Empatica E4 device

```

import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load EDA and HR data
eda_df = pd.read_csv("C:/Users/gunja/Desktop/Tecnomers Internship/systematic lit review/WESAD DATA analysis/S4/EDA.csv", skiprows=2, header=None, names=["EDA"])
hr_df = pd.read_csv("C:/Users/gunja/Desktop/Tecnomers Internship/systematic lit review/WESAD DATA analysis/S4/HR.csv", skiprows=2, header=None, names=["HR"])

# == 2. Define window size (e.g., 60 samples per window) ==
window_size = 60 # assuming 4 Hz sampling for EDA, ~1 Hz for HR
num_windows = min(len(eda_df), len(hr_df)) // window_size

features = []
labels = []

```

```

# === 3. Extract statistical features per window ===
for i in range(num_windows):
    start = i * window_size
    end = start + window_size

    eda_window = eda_df["EDA"][start:end]
    hr_window = hr_df["HR"][start:end]

    if len(eda_window) < window_size or len(hr_window) < window_size:
        continue

    feature_vector = [
        eda_window.mean(),
        eda_window.std(),
        hr_window.mean(),
        hr_window.std()
    ]
    features.append(feature_vector)

    # Simulated labels: 0 = rest, 1 = mental task, 2 = physical
    # Repeat pattern or assign randomly for demonstration
    labels.append(i % 3)

# === 4. Convert to DataFrame ===
X = pd.DataFrame(features, columns=["EDA_mean", "EDA_std", "HR_mean", "HR_std"])
y = np.array(labels)

# === 5. Split data ===
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

```

[11] ✓ 1.3s

```

# === 6. Train model (choose one) ===

# Random Forest
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
y_pred = rf_model.predict(X_test)

# XGBoost (uncomment if using this instead)
#xgb_model = XGBClassifier(use_label_encoder=False, eval_metric='mlogloss', random_state=42)
#xgb_model.fit(X_train, y_train)
#y_pred = xgb_model.predict(X_test)

# === 7. Evaluate ===
print("Classification Report:")
print(classification_report(y_test, y_pred))

print("Confusion Matrix:")
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=["Rest", "Mental", "Physical"], yticklabels=["Rest", "Mental", "Physical"])
plt.xlabel("Predicted")
plt.ylabel("True")
plt.title("Activity Type Detection Confusion Matrix")
plt.show()

# === 8. Feature Importance ===
importances = rf_model.feature_importances_
plt.figure(figsize=(6,4))
sns.barplot(x=X.columns, y=importances)
plt.title("Feature Importances (Random Forest)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```

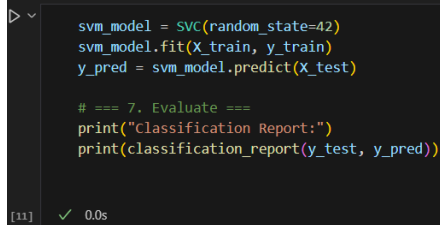
[14] ✓ 0.3s

Fig 3.15 Personalized Activity Type Detection using Random forest and XGboost

The above code is used to build a personalized activity type detection model using XGboost and random forest. For this the EDA and HR csv file is imported, then its is split into training and test sets, 20% of the data goes to testing, 80% to training. The random_state ensures we get the same

split every time. The training phase the data was used to teach the model to recognise the patterns using random forest classifier, the fit function was used to fit the data. The classifier was configured with 100 decision tree `n_estimators=100`.

The model was tested by making prediction on the test data using the predict function which were given the labels (`y_pred`) were then compared to the true labels (`y_test`) using the `classification_report()` from scikit-learn. This report provides key evaluation metrics such as **precision**, **recall**, **F1-score**, and **support** for each class, helping to understand how well the model performs across different activity types.



```
> ~
svm_model = SVC(random_state=42)
svm_model.fit(X_train, y_train)
y_pred = svm_model.predict(X_test)

# === 7. Evaluate ===
print("Classification Report:")
print(classification_report(y_test, y_pred))

[11] ✓ 0.0s
```

Fig 3.16 Personalized Activity Type Detection using Support vector machine

Alternatively, an **XGBoost Classifier** and **support vector machine** was also implemented in the code in place of the Random Forest.

CHAPTER 4

FINDINGS & SUGGESTIONS

4.1 FINDINGS

The systematic literature review revealed that the use of machine learning algorithms for heart disease detection using wearable devices has gained significant traction in recent years. Out of 884 research articles screened, 18 were selected for detailed analysis. Among these, the most widely used machine learning algorithms included Random Forest, XGBoost, K-Means Clustering, and Convolutional Neural Networks—all demonstrating strong accuracy in detecting and diagnosing heart disease. As part of the bibliometric analysis, insights were generated through abstract term analysis, keyword analysis, co-occurrence of author keywords, and both related and unrelated co-authorship analysis.

An exploratory analysis was also conducted on the WESAD respiratory data of patient no. 2. The respiration signal analysis revealed normal chest wall movements indicating inspiration and expiration, along with an episode of deep tidal breathing. Overall, the segment reflects a calm respiratory state, making it suitable for baseline or neutral affective analysis.

Insight from Recurrence plot Analysis of complete respiratory signal

RECURRENCE PLOT GRAPH AND RESPIRATION ANALYSIS

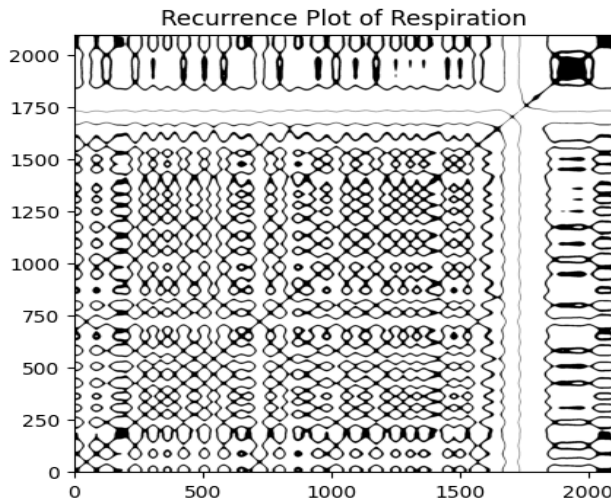


Fig 4.1 Recurrence plot

breathing_rate_bpm: 21.26
 mean_amplitude: 19546.81
 amplitude_std: 1717.36
 mean_bbi_sec: 2.82
 bbi_std: 0.78
 numBreathsDetected: 2249

To conduct a detailed analysis of the respiration signal using data extracted from the RespiBAN device, the signal underwent preprocessing that included centering, down-sampling to 35 Hz, and bandpass filtering (0.05–0.8 Hz) to isolate the physiological breathing component.

From the filtered signal, several respiratory features were extracted:

- The average breathing rate was **21.26 breaths per minute**, indicating a **slightly elevated respiratory rhythm**. (normal respiratory rhythm 12 to 18 breaths per minute)
- The mean amplitude of the respiration cycles was **19,546.81 (ADC units)** with a standard deviation of **1,717.36**, suggesting **strong and moderately variable breath depth**.
- The mean breath-to-breath interval was **2.82 seconds**, with a variability of **0.78 seconds**, indicating **mild irregularity** in timing.

A 20-second window of the respiration signal was used to generate a recurrence plot, revealing a **highly structured and self-similar pattern**. The **recurrence plot** exhibited **dense diagonal line structures** and a **regular grid-like pattern**, characteristic of **periodic and predictable dynamics**

in the breathing signal. This was quantitatively supported by a **Recurrence Rate of 0.20** and a **Determinism value of 1.0**, indicating that **100% of the recurrent points were part of predictable sequences**.

Taken together, these findings suggest that the subject was in a **non-stressed, physiologically stable state**, likely experiencing **mild arousal or engagement**, such as during amusement or focused attention. The respiratory behaviour was consistent, rhythmic, and structured, with no indication of irregular or chaotic breathing patterns typically associated with acute stress responses.

Insight from the Comparison of Breathing Features by Segment

RESPIBAN PARAMETER GRAPH

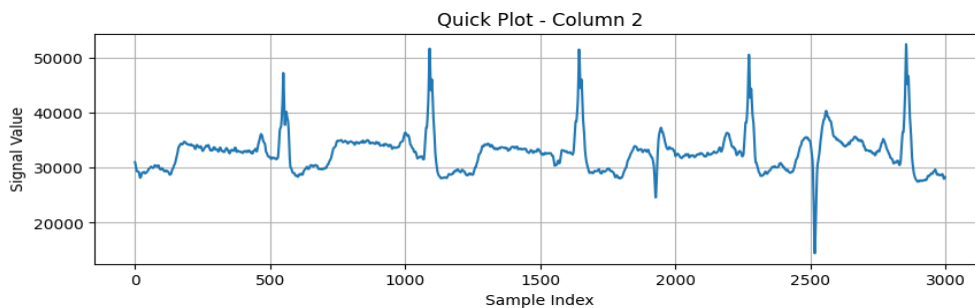


Fig 4.2 ECG

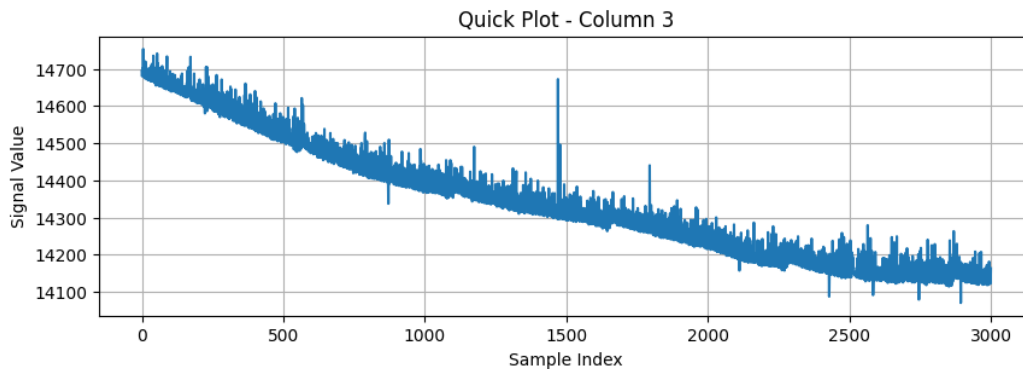


Fig 4.3 EDA

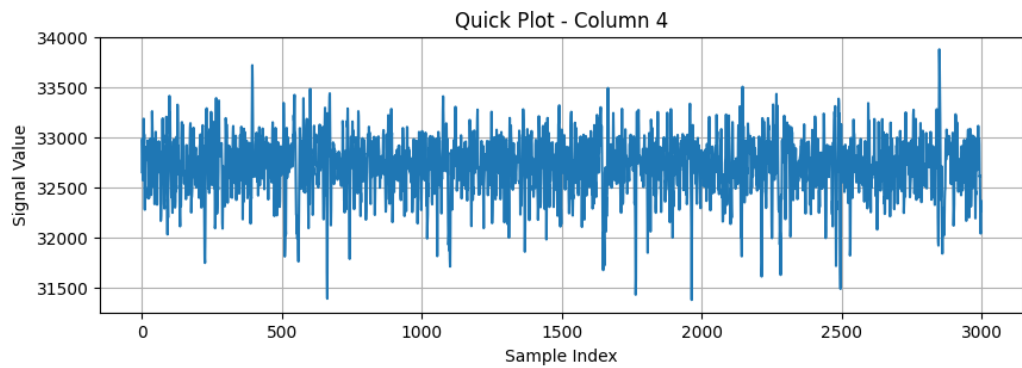


Fig 4.4 EMG

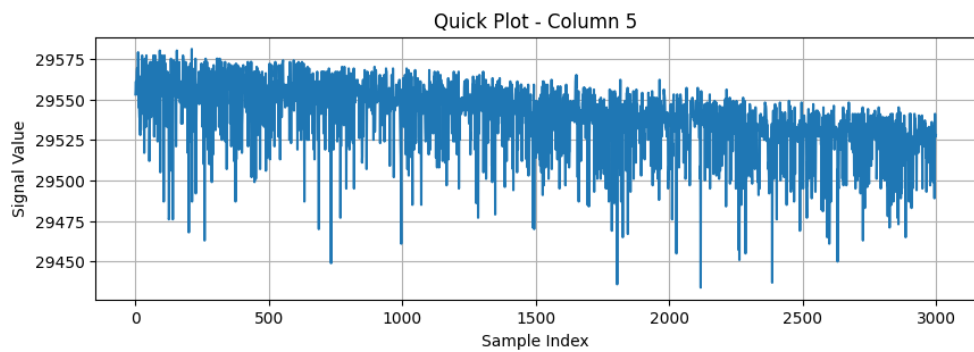


Fig 4.5 TEMP

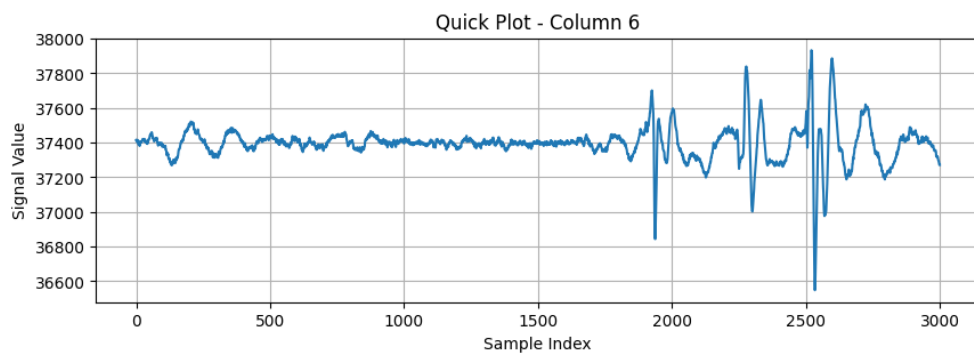


Fig 4.6 XYZ

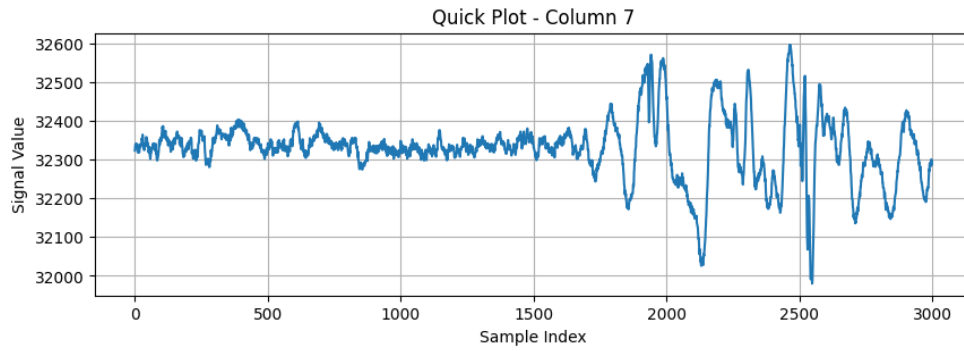


Fig 4.7 XYZ

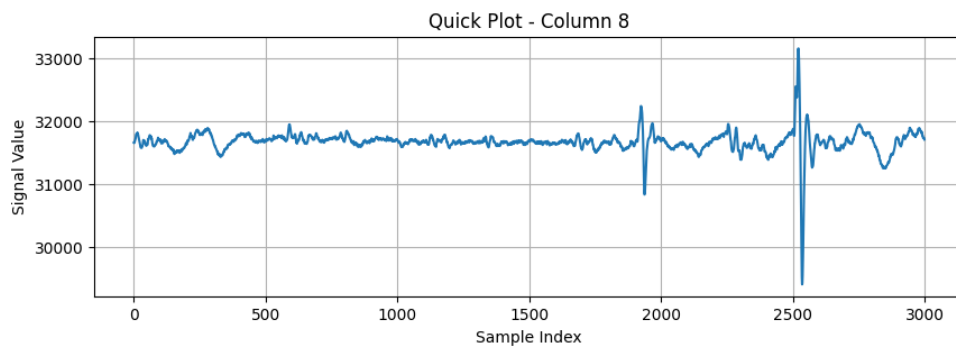


Fig 4.8 XYZ

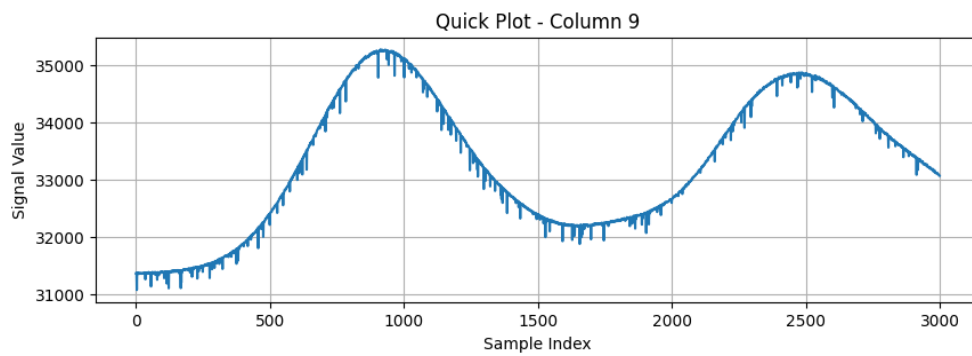


Fig 4.9 RESPIRATORY

DOWNSIZED RESPIRATORY SIGNAL ANALYSIS AND THE INSIGHT FROM IT

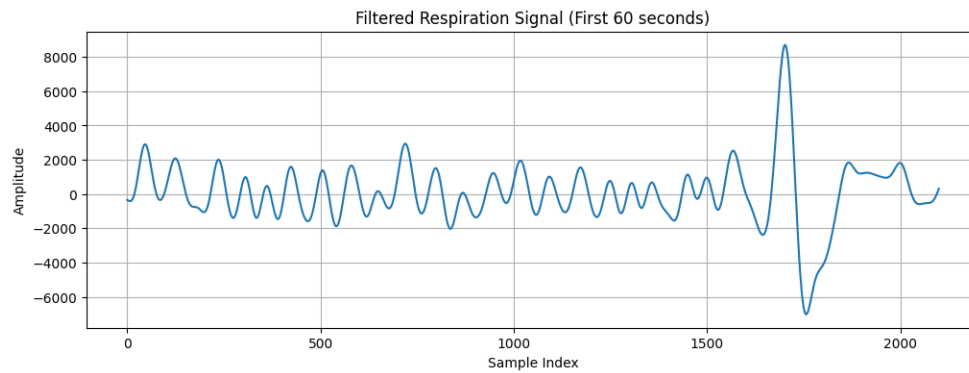


Fig 4.10 DOWNSIZED RESPIRATORY SIGNAL

The filtered respiration signal over the first 60 seconds exhibits a consistent and rhythmic pattern, indicative of stable breathing. The waveform shows well-defined inhale-exhale cycles, supporting the reliability of the signal for physiological interpretation. A significant deviation appears briefly around the 45–50 second mark, likely representing a deep inhalation or brief artifact. Overall, the segment reflects a calm respiratory state suitable for baseline or neutral affective analysis.

Personalized Activity Type Detection

- **Random forest method**

```

Classification Report:

```

	precision	recall	f1-score	support
0	0.29	0.17	0.21	12
1	0.17	0.12	0.14	8
2	0.07	0.14	0.10	7
accuracy			0.15	27
macro avg	0.17	0.14	0.15	27
weighted avg	0.19	0.15	0.16	27

Fig 4.11 RANDOM FOREST CLASSIFICATION REPORT

The **classification report** of **random forest** method reveals that the model performs poorly across all three activity types, with an overall accuracy of just 15%. The precision, recall, and F1-scores are low for every class, indicating that the model struggles both to correctly identify each class and to avoid misclassifications. This suggests that either the features used (mean and standard deviation

of EDA and HR) are not sufficient to distinguish between Rest, Mental, and Physical activities, or the amount and quality of the data are not adequate for training a robust model. Where each row (0, 1, 2) represents a class:

- 0 → Rest
- 1 → Mental
- 2 → Physical

Each column represents:

- **Precision:** How many predicted were correct?
- **Recall:** How many actual were correctly predicted?
- **F1-score:** Harmonic mean of precision & recall.
- **Support:** Number of actual instances per class.

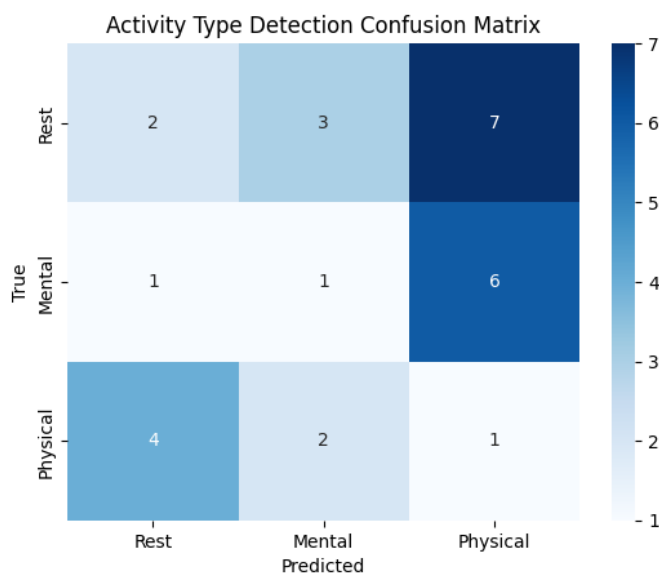


Fig 4.12 RANDOM FOREST CONFUSION MATRIX

The confusion matrix indicates that the model has difficulty accurately distinguishing between activity types, particularly between Rest and Physical activities. The high rate of misclassification suggests that the features used—mean and standard deviation of EDA and HR—may not adequately capture the physiological differences between these activity states. This limitation is likely due to the insufficient amount of data available for analysis. Future improvements could include extracting more complex features (e.g., peak detection, frequency-domain metrics) and enhancing data quality

and quantity.quantity of labelled data.

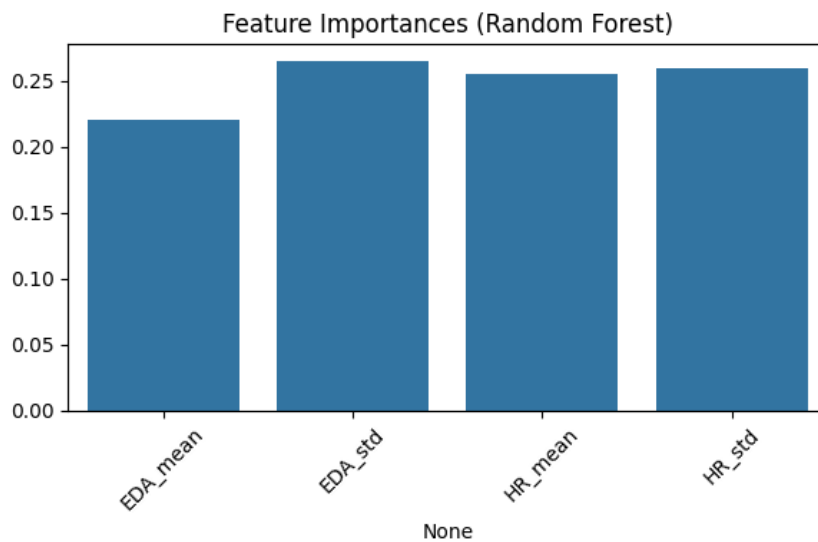


Fig 4.13 RANDOM FOREST FEATURE IMPORTANCE GRAPH

Feature importance indicates how useful each feature was in making decisions (splits) within the trees of the Random Forest model. The higher the value, the more the feature contributed to improving the model's predictions.

The **feature importance analysis** from the Random Forest model reveals that all four physiological features—EDA mean, EDA standard deviation, HR mean, and HR standard deviation—play meaningful roles in activity type classification. Among them, the standard deviation of EDA is the most significant, indicating that variability in skin conductance is a strong indicator of activity. Heart rate features also contribute substantially, particularly the standard deviation, reflecting how heart rate variability can distinguish between rest, mental, and physical activities. This insight can guide feature selection in future models to improve classification accuracy.

- **XGboost method**

Classification Report:					
	precision	recall	f1-score	support	
0	0.33	0.17	0.22	12	
1	0.27	0.38	0.32	8	
2	0.20	0.29	0.24	7	
accuracy			0.26	27	
macro avg	0.27	0.28	0.26	27	
weighted avg	0.28	0.26	0.25	27	

Fig 4.14 XGBOOST CLASSIFICATION REPORT

This classification report shows that the model achieved an overall accuracy of 26%, indicating limited predictive performance. Among the three classes, Class 1 had the highest recall (0.38), while Class 0 had the highest precision (0.33), though both values are relatively low. The macro and weighted F1-scores, both around 0.25–0.26, further confirm that the model struggled to accurately classify the data. This suggests issues such as class imbalance, insufficient features, or underfitting, and highlights the need for improved feature engineering and model tuning.

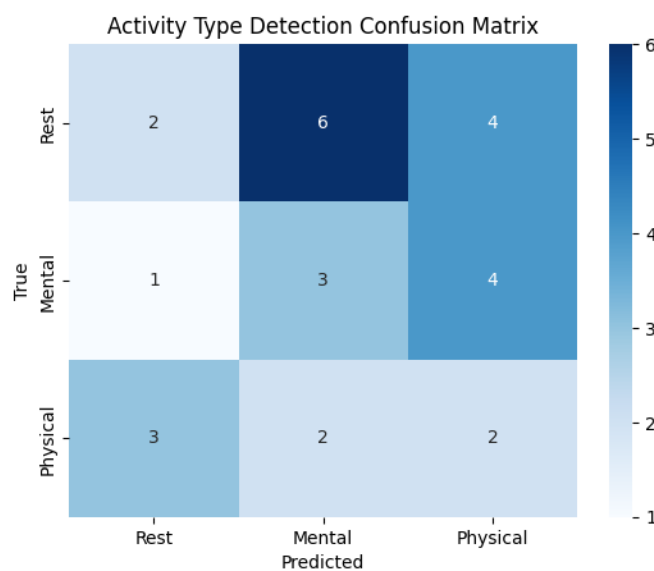


Fig 4.15 XGBOOST CONFUSION MATRIX

This confusion matrix shows the performance of a machine learning model in classifying three activity types: Rest, Mental, and Physical. The model struggles with clear differentiation between classes. For example, out of 12 actual "Rest" samples, only 2 were correctly classified, while 6 were misclassified as "Mental" and 4 as "Physical." Similarly, "Mental" activity was predicted correctly only 3 times out of 8, with notable confusion toward "Physical." "Physical" activity was also often misclassified, with only 2 correct predictions out of 7. Overall, the model shows significant overlap and confusion across classes, indicating the need for improved feature selection, model tuning, or better data representation to enhance classification accuracy.

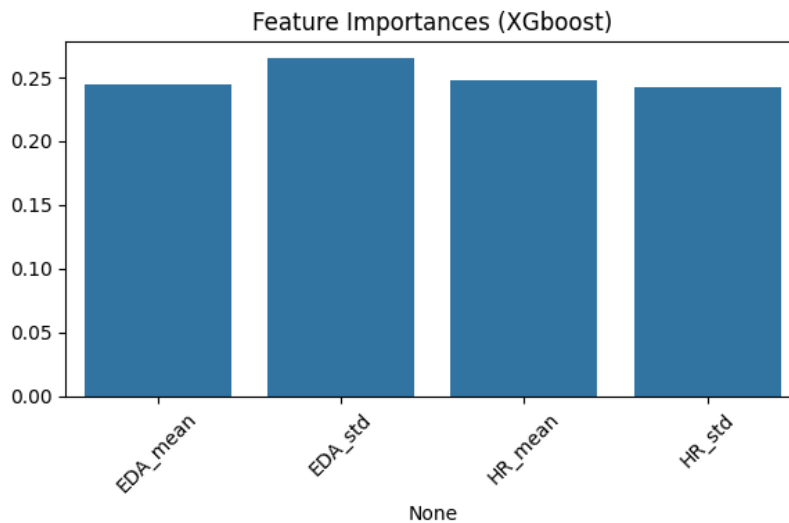


Fig 4.16 XGBOOST FEATURE IMPORTANCE

Among EDA mean, EDA standard deviation, HR mean, and HR standard deviation, the standard deviation of EDA is the most significant, indicating that variability in skin conductance is a strong indicator of activity. Heart rate features also contribute substantially, particularly the mean, also play a critical role in distinguishing activities even though the pivotal role is of EDA_std

- **Support Vector Machine method**

Classification Report:				
	precision	recall	f1-score	support
0	0.00	0.00	0.00	12
1	0.00	0.00	0.00	8
2	0.26	1.00	0.41	7
accuracy			0.26	27
macro avg	0.09	0.33	0.14	27
weighted avg	0.07	0.26	0.11	27

Fig 4.17 SVM CLASSIFICATION REPORT

This classification report shows that the model performed poorly in classifying the three activity types. It achieved an overall accuracy of only **26%**, with **class 2** (likely "Physical") being the only class it could predict with some success—achieving a **recall of 1.00** but with low **precision (0.26)**. Classes 0 and 1 (likely "Rest" and "Mental") had **zero precision, recall, and F1-score**, indicating the model failed to identify any samples from those classes correctly. The low **macro and weighted average F1-scores (0.14 and 0.11)** confirm this imbalance.

This suggests the model is heavily biased toward class 2 and unable to generalize across all classes, pointing to potential issues like class imbalance, insufficient feature quality, or underfitting.

Table 4.1 MODEL COMPARISION TABLE

Metric	SVM	XGBoost	Random Forest
Accuracy	0.26	0.26	0.15
Macro Avg Precision	0.09	0.27	0.17
Macro Avg Recall	0.33	0.28	0.14
Macro Avg F1-Score	0.14	0.26	0.15
Weighted Avg Precision	0.07	0.28	0.19
Weighted Avg Recall	0.26	0.26	0.15
Weighted Avg F1-Score	0.11	0.25	0.16

- **XGBoost outperformed** both SVM and Random Forest across nearly all metrics, showing better **precision, recall, and F1-scores**, especially in macro and weighted averages.
- **SVM** showed slightly better recall (0.33 macro avg), but had very poor precision and overall F1-score, indicating it predicted one class correctly but failed at distinguishing between all classes.
- **Random Forest** had the **lowest accuracy (0.15)** and performed poorly across all metrics, suggesting it struggled with class separation in this dataset.

Overall, **XGBoost is the most balanced and effective model** among the three for this classification of this unstructured data. Unstructured data—such as physiological signals collected through wearable sensors—often lacks consistent patterns, contains noise, and may have high variability across individuals. Given this complexity and the relatively small dataset size, models like Random Forest, XGBoost, and SVM struggle to learn meaningful patterns and generalize effectively. Limitation is described in detail in chapter 5.

4.2 SUGGESTIONS

By leveraging the rich, continuous physiological data captured by wearable devices, sophisticated machine learning algorithms can be employed for enhanced data acquisition through multi-sensor fusion, noise reduction, personalized baseline establishment, and advanced feature extraction. This enables the development of robust predictive models for early cardiovascular detection, cardiac event risk stratification, heart failure monitoring, and post-event rehabilitation.

Ultimately, this technology can be enhanced and integrated further to facilitate personalized and proactive healthcare through tailored alert systems, remote patient monitoring, seamless EHR integration, and data-driven lifestyle recommendations. Moreover, the data required for conducting through machine learning analysis was incomplete so it was difficult to conduct a thorough analysis. The current analysis was conducted using the most optimal techniques identified through the systematic review of research articles from the Scopus database.

The current model using basic EDA and HR features shows low accuracy in detecting activity types. To improve performance, more informative features (like signal peaks or entropy) should be extracted. Using larger or overlapping time windows, handling class imbalance, and possibly trying advanced models like LSTM or CNN can further enhance accuracy. Adding physiological context can also help the model differentiate between Rest, Mental, and Physical activities. However, for a thorough analysis, to the reviewed data necessary adjustments has to be made along with careful consideration of data privacy and security, algorithm transparency, bias mitigation, rigorous clinical validation, and effective user education to ensure responsible and impactful translation into clinical practice and to ensure the best result.

CHAPTER 5

CONCLUSION

The internship at Technomers, was a valuable opportunity to gain firsthand experience in understanding and analyzing the practical challenges of use of analytics within an organization. The primary objective of studying and providing the clients solutions using latest technology was successfully achieved through detailed observations, interactions, and systematic literature analysis.

Throughout the internship, several gaps were identified, in the finding the most suitable solution for the requested problem. The problem was categorized and the solution was searched from different database for selection of the most apt database for getting the solutions for client. By applying theoretical concepts learned in the classroom, combined with practical insights from the workplace, solutions were proposed. The project helped in recommendation decision to Technomers client Movano for better decision making in wearable device using machine learning, but also significantly enhanced the intern's understanding of business analytics and its use for enhancing client requirements and problem solving. The overall experience reinforced the value of technology and data-driven decision-making in optimizing business operations.

5.1 LIMITATIONS

During the course of the internship, a structured analysis was conducted using the collected data to understand the nature of the problem and explore potential machine learning solutions. Although a complete predictive model was not developed, the data was preprocessed and evaluated across multiple machine learning techniques to assess their suitability for the client's use case. The project concluded from the literature analysis that algorithms such as Random Forest, XGBoost based on factors like performance metrics, interpretability, are applicable to the client's scenario.

However, few limitations that were encountered during this process. The dataset used in this analysis is unstructured and exploratory in nature, which significantly affects the performance of traditional machine learning models. The performance metrics observed across these models (e.g.,

low precision, recall, and accuracy) indicate that they are not well-suited to handle such irregular, sparse, or under-labeled data in their current form. This highlights the need for more advanced learning strategies and robust data preprocessing to improve classification outcomes. Furthermore, the lack of domain-specific annotations in the dataset posed challenges in accurately assessing algorithm effectiveness for context-sensitive tasks. Future work would benefit from a more comprehensive dataset, real-time testing environments, and deeper feature engineering to develop and evaluate robust models tailored to client needs.

5.2 COURSE OF ACTION

To address the identified limitations, several steps can be taken to improve the quality and impact of future work. First, increasing the volume and diversity of the dataset will be critical—this may involve collecting more samples from a wider population and ensuring balanced representation across different conditions or user segments. Secondly, enriching the dataset with clearer, domain-specific annotations (e.g., labeled emotional states or activity types) will allow for more accurate evaluation and training of future models.

To enhance model performance in future iterations, particularly with limited and unstructured data, one-shot learning and few-shot learning techniques can be employed. These approaches allow a model to learn from very few labeled examples—ideal for healthcare data where labeled samples are often scarce or expensive to obtain. Additionally, applying fine-tuning and reinforcement learning techniques can help models adapt more effectively to subtle variations in the input signals. Reinforcement learning, in particular, can be used to optimize decision-making over time by learning from feedback loops. When paired with larger sample sizes, more balanced data, and advanced signal preprocessing, these approaches can significantly improve the robustness and accuracy of machine learning models in handling complex biomedical data collected through wearable technologies.

Finally, collaboration with domain experts (e.g., healthcare professionals or wearable tech developers) can ensure that the chosen approach is both technically sound and contextually relevant. By implementing these steps, future work can progress from theoretical evaluation to a

robust, deployable solution tailored to the client's specific needs.

5.3 LEARNING

The internship provided me with significant technical and professional learning experiences. On the technical front, I gained hands-on experience in Python programming, SQL, and data visualization tools like Seaborn and Matplotlib. I learned to clean and preprocess real-world physiological data collected from wearable devices and applied machine learning models such as Random Forest and XGBoost to classify health-related states. Through exploratory data analysis and recurrence plotting, I understood how internal bio-signals like EDA and heart rate can be used to detect stress and activity types. Additionally, I conducted a systematic literature review using Scopus and performed bibliometric analysis using VOSviewer, which enhanced my understanding of research trends in health-tech and machine learning.

Beyond technical skills, the internship sharpened my analytical thinking and problem-solving abilities in a real-world context. I improved my communication skills by presenting findings to both academic and industry mentors and developed better time management and task prioritization while working independently. Exposure to the fast-paced environment of a tech startup gave me insight into the dynamics of product development and client-focused solution building. Furthermore, I recognized the importance of ethical considerations such as data privacy, algorithm transparency, and responsible use of AI in healthcare applications. Overall, this experience bridged the gap between academic knowledge and industry expectations, strengthening both my technical foundation and professional mindset.

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